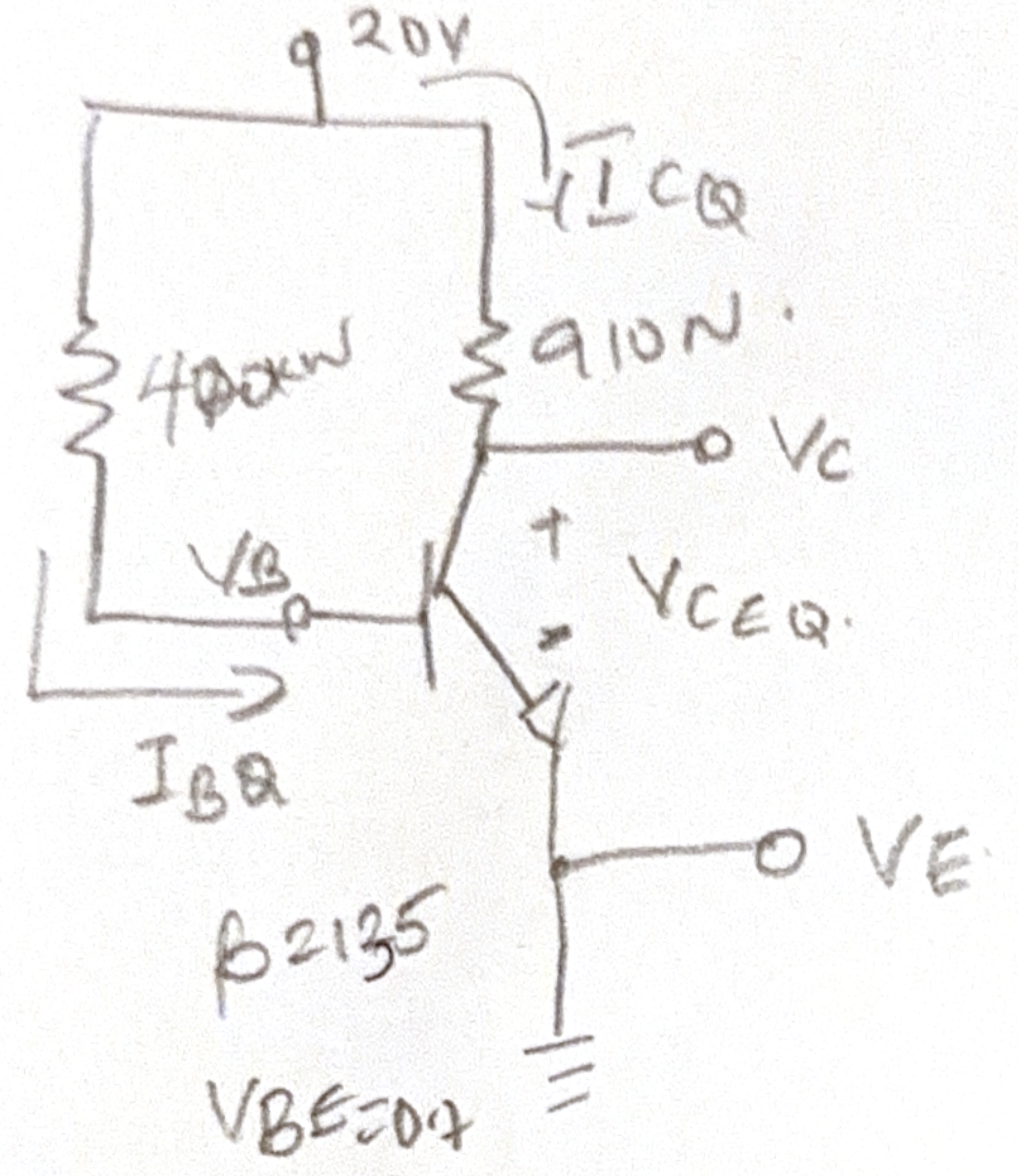
Name! Naimot Yekini

(1B)




$I_C = \beta I_B$

$I_E = I_C + I_B$

$I_E = (\beta + 1) I_B$

$I_E \approx I_C$.

$V_{CC} = 20V$

$R_B = 470k\Omega$.

$R_C = 910\Omega$.

$\beta = 135$

$V_{BE} = 0.7V$

Emitter is grounded = 0.

① $I_B = \dfrac{V_{CC} - V_{BE}}{R_B}$

$I_B = \dfrac{20 - 0.7}{470000} = \dfrac{19.3}{470000} =$

$4.106 \times 10^{-5} A$

$I_B = 41\mu A$

② $I_C = I_C = \beta I_B$

$I_C = 135 \times 41\mu A$

$I_C = 5.54 mA$

③ $I_E = I_C + I_B$

$I_E = 5.54 \times 0.041 mA$

$I_E = 5.58 mA$

④ $V_C$ $V_{RC} = I_C R_C$

$V_{RC} = 5.54 mA \times 910\Omega = 5.04V$

$V_C = V_{CC} - V_{RC}$

$V_C = 20 - 5.04 = 14.96V$.

$V_B = VB = VBE = 0.7V.$

6) $V_E = VE = 0V$

$V_{CE} = V_C - V_E$

$V_{CE} = 14.96 - 0$

$= 14.96V$

Qpoint = $(V_{CE} = 15V,\ I_C = 5.5mA.$